

# Sam Watts

## Machine Learning Engineer

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I'm a data professional with a wide range of experience in different types of modelling and software engineering practices. I originally specialised in recommender systems, but more recently I have followed my interest in other topics, including: geospatial data, computer vision, and agentic systems.

I'm driven by a desire to learn, solve important problems, and be helpful. I'm most interested in roles in climate tech and health; and I'm open to other roles that are socially useful.

I'm currently based in Edinburgh, Scotland

I'm currently looking for in-person roles in Edinburgh, fully remote roles, or hybrid roles in the UK with occasional travel.

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## Experience

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### Freelance

**Software Engineer • Sep 2026 — Now**

Creating ML coding tasks used as RLHF training data for agentic coding models.

### GreenFlip

**Founding ML Engineer • Sep 2025 — Sep 2026 • 1yr**

I built ML models to assess residential homes for low-carbon heating retrofit, working across computer vision, geospatial, and tabular modelling. My solar panel placement algorithm had higher accuracy on maximum placeable panels than the Google Solar API on our internal evaluation set of diverse UK homes.

### Continuum Industries

**Senior Software Engineer • May 2024 — Aug 2025 • 1yr 4m**

I worked with genetic algorithms, machine learning and geospatial data to create tools that speed up the design of large-scale linear infrastructure projects - most notably electricity transmission lines, whose fast development is crucial for the transition to renewable energy.

- Led improvements to genetic algorithms, including smart visualisation of spatially diverse routes and efficient parallel runs of the core algorithms
- Developed 2 ML prototypes to extract data from analogue maps and measure the width of roads from satellite imagery
- Developed LLM-based tools for analysing possible transmission routes and generating technical reports

### University of Edinburgh

**Research Fellow • Aug 2023 — Mar 2024 • 8m**

I worked with a novel retinal image dataset to develop deep learning models for the early detection of age-related macular degeneration, glaucoma and other retinal diseases.

- Implemented semantic segmentation and classification models in PyTorch
- Worked with retinal imaging foundation models using ViT architectures

### Too Good To Go

**Machine Learning Engineer • Oct 2021 — Mar 2023 • 1yr 6m**

I worked on building a product recommendation system from the ground up. My work spanned a range of ML areas, from EDA and model development to CI/CD and productionising model workflows.

- Designed and implemented a recommender model that increased user purchase conversion by 0.6pp versus a baseline in A/B tests (projected value of £1m+ per year)
- Led the adoption of Metaflow as our ML workflow tool, significantly speeding up development
- Experimented with new modelling approaches inspired by research presented at RecSys 2022

### John Lewis Partnership

**Lead Data Scientist, Product Recommendations • Mar 2021 — Oct 2021 • 8m**

Led a team of 4 in the design and execution of data science and data engineering work, focusing on development of existing and new recommendations models - whilst still writing production code. Achievements:

- Implemented hyperparameter tuning using sklearn pipelines across all in-house models - boosting recommendation performance in A/B tests
- Iterated on a sequence learning (Prod2Vec) model built on web browsing data in TensorFlow by exposing extra hyperparameters

### ***Data Scientist, Product Recommendations • Aug 2020 — Mar 2021 • 8m***

Responsible for developing in-house recommendation algorithms and ensuring they delivered good results in production. Achievements:

- Productionised a previously tested model on GCP, using AI Platform for training and serving
- Drove implementation of a solution for more granular analysis of A/B tests using Airflow and BigQuery
- Organised a regular journal club session, where we discussed recent publications and used them to inform our product roadmap

### ***Data Scientist • Apr 2018 — Aug 2020 • 2yrs 5m***

Worked on a variety of different projects as one of the first members of a new Data Science team, and gained exposure to a number of different business areas and modelling techniques, including:

- Product demand forecasting
- Supply chain optimisation using vehicle routing models
- Warehouse process simulation and optimisation
- Evaluation of experiments using counterfactual methods

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## **Education**

### **University College London (UCL)**

*Natural Sciences, BSc - Organic Chemistry and Neuroscience • 2013 — 2017*

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## **Projects**

### **Carbon13 Venture Builder Programme**

*Venture Founder • 2025*

I explored multiple areas of climate tech from both a business and technical perspective. With a co-founder, I worked on validating software tooling for recipe formulation in the food industry, with a focus on novel ingredients from precision fermentation and cultivated meat. We ultimately decided not to continue the venture together, but it was a valuable crash course in early-stage company building.

### **DataKind**

*DataDive Volunteer & Data Ambassador • 2021 - 2025*

I have volunteered on multiple DataDives - which help charities extract value from their data using data science techniques. I also helped to organise several of these events as a Data Ambassador - by framing key analytical questions, cleaning and documenting data, and liaising with charity representatives to enable volunteers to develop good solutions.

### **Saturdays.AI Copenhagen**

*Volunteer Mentor • 2022*

Helped teach data science & machine learning foundations to a group of beginners. I taught two sessions, on Exploratory Data Analysis and Clustering.

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## **Tools**

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|-------------------------------------------------------|-------------------------------------------|
| • Python                                              | • Docker                                  |
| • ML libraries - sklearn, tensorflow, pytorch, pandas | • CI/CD - Gitlab, Jenkins, Github Actions |
| • SQL - Redshift, Bigquery, MSSQL, Teradata, dbt      | • ML Workflows - Metaflow, AWS Sagemaker  |
| • Cloud Platforms - AWS & Google Cloud                | • Terraform                               |
| • Git                                                 | • Airflow                                 |